1. PROJECT TITLE :-

*** Heart Disease Analysis***

PROJECT NAME: **HEART DISEASE**
TEAM LEAD: **KONDAGORRI RAVI**
TEAM MEMBERS: **1.K RAVI**
2.B JOY
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TECHNOLOGY: **1.DATA ANALYSIS**
2.VISUALIZATION
3.TABLEAU (BUSINESS INTELLIGENCE SOFTWARE)

2. PROBLEM STATEMENT

Heart disease remains one of the leading causes of mortality worldwide, with its prevalence increasing due to lifestyle changes, poor dietary habits, and lack of physical activity. Despite medical advances, prevention and early detection are critical to reducing severe outcomes. This project focuses on analyzing heart disease data to identify risk factors and propose data-driven strategies for awareness and prevention.

3. DATA COLLECTION

Dataset: Heart Disease Dataset Records: 300+ patients Columns: Age, Gender, Cholesterol, Blood Pressure, Exercise Frequency, Smoking Status, Diabetes, Family History, ECG Results, etc.

4. DATA PREPARATION

- **Removed null and duplicate values**
- **Created calculated fields: Age Groups, Cholesterol Levels (bins), Risk Score Index**
- **Grouped data by demographic factors and clinical indicators**

5. DATA VISUALIZATION

10 charts created:

- **Bar: Gender vs Heart Disease Cases**
- **Scatter: Cholesterol vs Age**
- **Bubble: Risk Factors by Frequency**
- **Line: Exercise vs Blood Pressure**
- **Stacked Bar: Smoking Status vs Disease Prevalence**
- **Heatmap: Correlation between Risk Indicators …and more**

6. DASHBOARD

Name: Heart Disease Risk Analytics Dashboard Filters: Age, Gender, Cholesterol Level, Smoking Status Tool: Tableau Link: https://public.tableau.com/app/profile/ravi.and.co.team/vizzes

7. STORY

Name: From Data to Diagnosis: A Heart Health Journey 5 Story Points with insights (e.g., lifestyle impact, cholesterol patterns, exercise benefits, smoking risks, prevention strategies)

Tableau Link https://public.tableau.com/app/profile/ravi.and.co.team/vizzes

8. KEY FINDINGS

- **65% of patients with cholesterol > 240 mg/dL showed higher risk of heart disease**
- **Regular exercise (3+ hrs/week) reduced risk scores by 20%**
- **Smokers had 30% higher prevalence compared to non-smokers**
- **Patients with family history + diabetes showed the highest combined risk**

9. RECOMMENDATIONS

1. **Awareness campaigns on cholesterol management**
2. **Free fitness programs for high-risk groups**
3. **Smoking cessation initiatives on campus/community**
4. **Preventive health check-ups for patients over 40**
5. **Nutrition counseling for diabetic patients**

10. CONCLUSION

Heart disease risk is strongly influenced by lifestyle and clinical factors. By leveraging Tableau dashboards and data storytelling, this project highlights actionable strategies for prevention. Implementing these recommendations can reduce risk prevalence by 15–20% and improve overall community health outcomes.